MTH240

Specimen Exam Paper Solutions

Engineering Mathematics I

Question 1

$$c_1\begin{pmatrix}-1\\1\\2\\0\end{pmatrix}+c_2\begin{pmatrix}2\\1\\0\\1\end{pmatrix}+c_3\begin{pmatrix}0\\1\\0\\-1\end{pmatrix}=\begin{pmatrix}1\\0\\-1\\1\end{pmatrix}$$

$$-c_1+2c_2=1$$
$$c_1+c_2+c_3=0$$
$$2c_1=-1$$
$$c_2-c_3=1$$

$$\left(\begin{array}{ccc|c}-1&2&0&1\\1&1&1&0\\2&0&0&-1\\0&1&-1&1\end{array}\right)\xrightarrow[R_3\to R_3+2R_1]{R_2\to R_2+R_1}\left(\begin{array}{ccc|c}-1&2&0&1\\0&3&1&1\\0&4&0&1\\0&1&-1&1\end{array}\right)\xrightarrow[R_3\to R_3-4R_4]{R_2\to R_2-3R_4}\left(\begin{array}{ccc|c}-1&2&0&1\\0&0&4&-2\\0&0&4&-3\\0&1&-1&1\end{array}\right)$$

$$\xrightarrow{R_2\leftrightarrow R_4}\left(\begin{array}{ccc|c}-1&2&0&1\\0&1&-1&1\\0&0&4&-3\\0&0&4&-2\end{array}\right)\xrightarrow{R_4\to R_4-R_3}\left(\begin{array}{ccc|c}-1&2&0&1\\0&1&-1&1\\0&0&4&-3\\0&0&0&1\end{array}\right)$$

The system has no solution hence $[1,0,-1,1]^T$ is not a linear combination of $[-1,1,2,0]^T$, $[2,1,0,1]^T$ and $[0,1,0,-1]^T$.

Question 2

$$A - \lambda I = \begin{pmatrix} 2-\lambda & 0 & 1 \\ -1 & 2-\lambda & 3 \\ 1 & 0 & 2-\lambda \end{pmatrix}$$
$$\det(A - \lambda I) = (2-\lambda)(\lambda^2 - 4\lambda + 3) = 0$$

$$\lambda = 1,2,3$$

For $\lambda_1 = 1$:

$$(A - \lambda I)\boldsymbol{v} = \mathbf{0}$$
$$\begin{pmatrix} 1 & 0 & 1 \\ -1 & 1 & 3 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$
$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ -1 & 1 & 3 & 0 \\ 1 & 0 & 1 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\boldsymbol{v} = \begin{pmatrix} -1 \\ -4 \\ 1 \end{pmatrix}$$

For $\lambda_2 = 2$:

$$(A - \lambda I)\boldsymbol{v} = \mathbf{0}$$
$$\begin{pmatrix} 0 & 0 & 1 \\ -1 & 0 & 3 \\ 1 & 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$
$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ -1 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$
$$\boldsymbol{v} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

For $\lambda_2 = 3$:

$$(A - \lambda I)\boldsymbol{v} = \mathbf{0}$$
$$\begin{pmatrix} -1 & 0 & 1 \\ -1 & -1 & 3 \\ 1 & 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$
$$\left(\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ -1 & -1 & 3 & 0 \\ 1 & 0 & -1 & 0 \end{array}\right) \to \left(\begin{array}{ccc|c} -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$
$$\boldsymbol{v} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

There are three linearly independent eigenvectors, M is diagonalizable.

Question 3

(a)

- $\mathbf{0} \in \boldsymbol{V}$ because $\mathbf{0} - \mathbf{0} + (\mathbf{2})(\mathbf{0}) = \mathbf{0}$.

- If $\boldsymbol{v}$ and $\boldsymbol{w} \in V$, then: $\boldsymbol{v_1} - \boldsymbol{v_2} + \mathbf{2}\boldsymbol{v_3} = \mathbf{0}$ and $\boldsymbol{w_1} - \boldsymbol{w_2} + \mathbf{2}\boldsymbol{w_3} = \mathbf{0}$

$$\rightarrow (\boldsymbol{v_1} + \boldsymbol{w_1}) - (\boldsymbol{v_2} + \boldsymbol{w_2}) + \mathbf{2}(\boldsymbol{v_3} + \boldsymbol{w_3}) = \mathbf{0}$$

  So, $\boldsymbol{v} + \boldsymbol{w} \in \boldsymbol{V}$.

- If $\boldsymbol{v} \in \boldsymbol{V}$, then: $\boldsymbol{v_1} - \boldsymbol{v_2} + \mathbf{2}\boldsymbol{v_3} = \mathbf{0} \rightarrow \boldsymbol{kv_1} - \boldsymbol{kv_2} + \boldsymbol{k}\mathbf{2}\boldsymbol{v_3} = \mathbf{0}$

  So, $\boldsymbol{kv} \in \boldsymbol{V}$ for every real number k.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y - 2z \\ y \\ z \end{bmatrix} = y\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$ are non-zero, non-parallel. They are linearly independent. Hence $\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}\right\}$ is a basis of V.

(b)

$$\begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 2 & 3 & 5 \end{pmatrix} \xrightarrow[R_3 - 2R_1]{R_2 + R_1} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 3 \\ 0 & 3 & 1 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 3 \\ 0 & 0 & -2 \end{pmatrix}$$

Now it's in row-echelon form. The non-zero rows form a basis for the row space:

$$\{[1 \quad 0 \quad 2], [0 \quad 3 \quad 3], [0 \quad 0 \quad -2]\}$$

To find the basis for the column space, use the original columns corresponding to the pivot columns in the row-echelon form.

So we take the corresponding columns from the original matrix A:

$$\left\{\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}\right\}$$

Question 4

(a) Let X represent the number of particles emitted during a 5-second period.

$$X \sim Poi(\lambda = 4)$$

$$P(X = 3) = \frac{e^{-4}.4^3}{3!} = 0.1954$$

(b)

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \frac{e^{-4}.4^0}{0!} = 0.9817$$

(c) Let Y represent the number of particles emitted during a 20-second period.

$$Y \sim Poi(\lambda = 16)$$

$$P(Y = 2) = \frac{e^{-16}.16^{10}}{10!} = 0.03409$$

(d) Let Z = number of interval out of 8 have exactly 3 particles emitted.

$$Z \sim Bi(n = 8, p = 0.1954)$$

$$P(Z = 5) = \binom{8}{5} 0.1954^5 (1 - 0.1954)^{8-3} = 0.0106$$

Question 5

(a) Let X represent the battery life of a smartphone: $X \sim N(6.5, 2.3^2)$

$$P(X < 5) = P\left(Z < \frac{5 - 6.5}{2.3}\right) = P(Z < -0.652) = 0.2578$$

Approximately **25.78%** of smartphones last less than 5 hours on a single charge.

(b)

$$P(4 < X < 10) = P\left(\frac{4 - 6.5}{2.3} < Z < \frac{10 - 6.5}{2.3}\right)$$
$$= P(-1.087 < Z < 1.522)$$
$$= 0.9357 - 0.1389 = 0.79680$$

The probability is approximately **79.68%**.

(c)

We need the value of X corresponding to the cumulative probability $P(Z < z) = 0.35$

Using a Z-table or inverse normal calculator, the Z-score for $P(Z < z) = 0.35$ is approximately $z = -0.385$.

Convert the Z-score back to X:

$$X = z * \sigma + \mu = -0.385 * 2.3 + 6.5 = 5.61$$

Answer: Approximately **5.61 hours** for 35% of smartphones to run out of charge.

----- END OF SOLUTIONS -----